1. N4: relation-collection index calculus on the GHS plane quartic vs a matched rho on E0(K)

2026-10-19. Branch `claude/quartic-ic-n4-20261019`, protocol `PROTOCOL.md` (commit 9a63c9161), records `results/`. Descriptive run report; interpretation in `README.md`. Measured values are in F_q-multiplication equivalents (M-eq); every IC run verified kG = Q on E0.

1.1. Result in brief

The old quartic "pure IC" enumerated the subgroup (T1). A real pipeline (line relations with one large prime, Lanczos mod ell, descent through canonical Pic0 arithmetic) verified on 51 of 51 instances and costs IC/rho = 0.122 [0.10, 0.16] at q = 262111. Exponent on q >= 4051: IC 1.083 [1.040, 1.129], rho 1.485 [1.437, 1.525], gap 0.402 [0.335, 0.468] (w = 0). Lead N4 stays open by the frozen rule. Scope: GHS-weak family only, plain rho, shared VM.

1.2. Cold single-target cost

q	units ok	rho runs	IC-B0 M-eq	rho M-eq	IC/rho w=0 (95%)	IC/rho w=1 (95%)	resolved
223	5/5	150	8.59e+05	2.04e+05	4.22 (3.55-4.92)	2.94 (2.55-3.36)	rho lower
499	5/5	150	1.59e+06	5.78e+05	2.75 (2.24-3.19)	1.97 (1.69-2.22)	rho lower
991	5/5	150	2.54e+06	1.51e+06	1.69 (1.49-1.91)	1.32 (1.19-1.45)	rho lower
2011	5/5	150	4.84e+06	4.48e+06	1.08 (0.97-1.21)	0.90 (0.82-0.99)	unresolved
4051	5/5	150	9.54e+06	1.3e+07	0.73 (0.66-0.82)	0.66 (0.60-0.72)	IC lower
8191	5/5	120	1.91e+07	4e+07	0.48 (0.44-0.52)	0.46 (0.42-0.50)	IC lower
16363	5/5	120	4.26e+07	1.11e+08	0.38 (0.35-0.43)	0.37 (0.34-0.41)	IC lower
32719	5/5	80	9.22e+07	2.76e+08	0.33 (0.30-0.37)	0.33 (0.30-0.37)	IC lower
65479	5/5	80	2e+08	9.04e+08	0.22 (0.20-0.25)	0.23 (0.20-0.26)	IC lower
131071	3/3	24	4.57e+08	2.45e+09	0.19 (0.12-0.29)	0.22 (0.14-0.35)	IC lower
262111	3/3	18	7.65e+08	6.28e+09	0.12 (0.10-0.16)	0.14 (0.12-0.19)	IC lower

Baseline arm IC-A0r (all rational points, retries charged): q=223: 2.04e+06 M-eq; q=499: 8.49e+06 M-eq; q=991: 3.12e+07 M-eq; q=2011: 2.24e+08 M-eq; q=4051: 5.03e+08 M-eq.

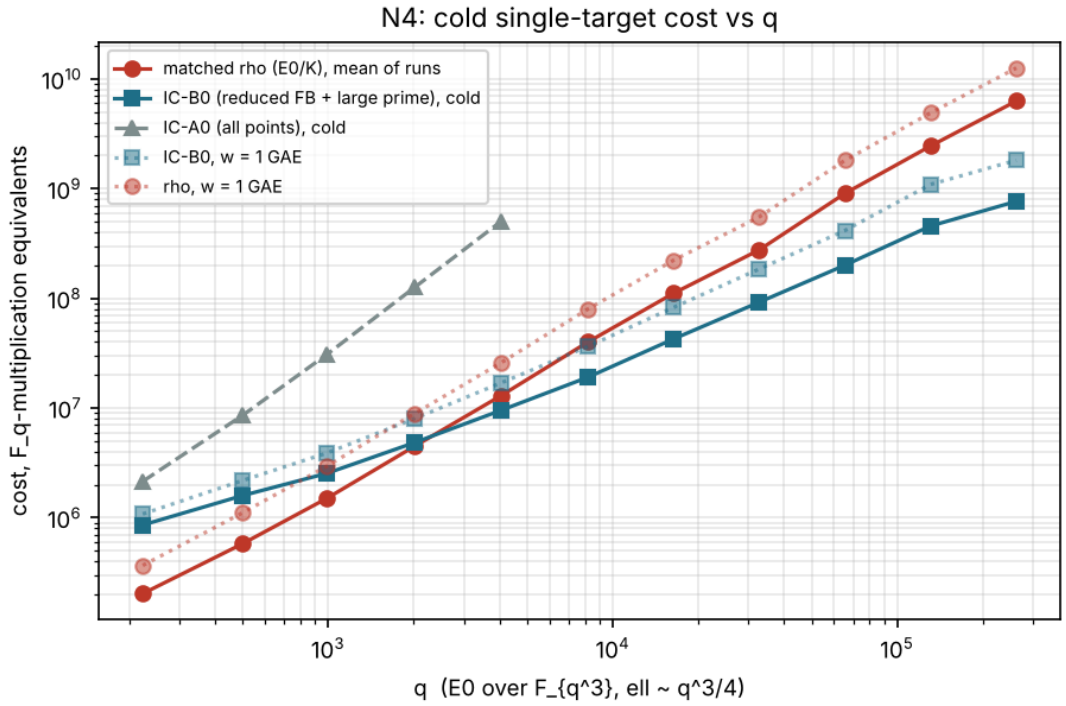


Figure 1: Cold cost vs q (recorded means; w = 1 GAE dotted). Source: results/summary_all.json, grid commit in README.

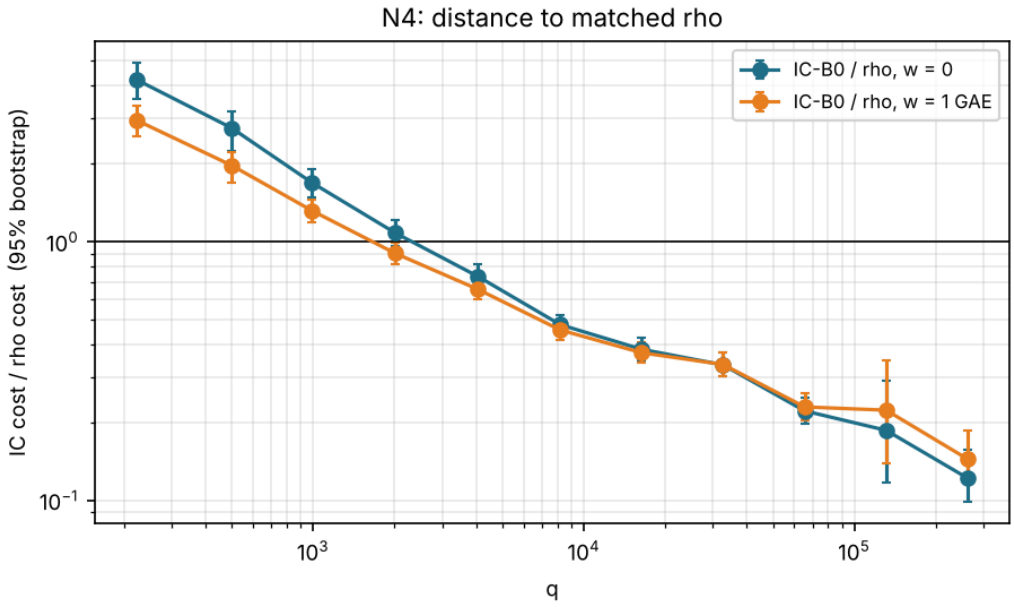


Figure 2: IC-B0 / rho with 95% bootstrap intervals (2000 resamples of units and runs).

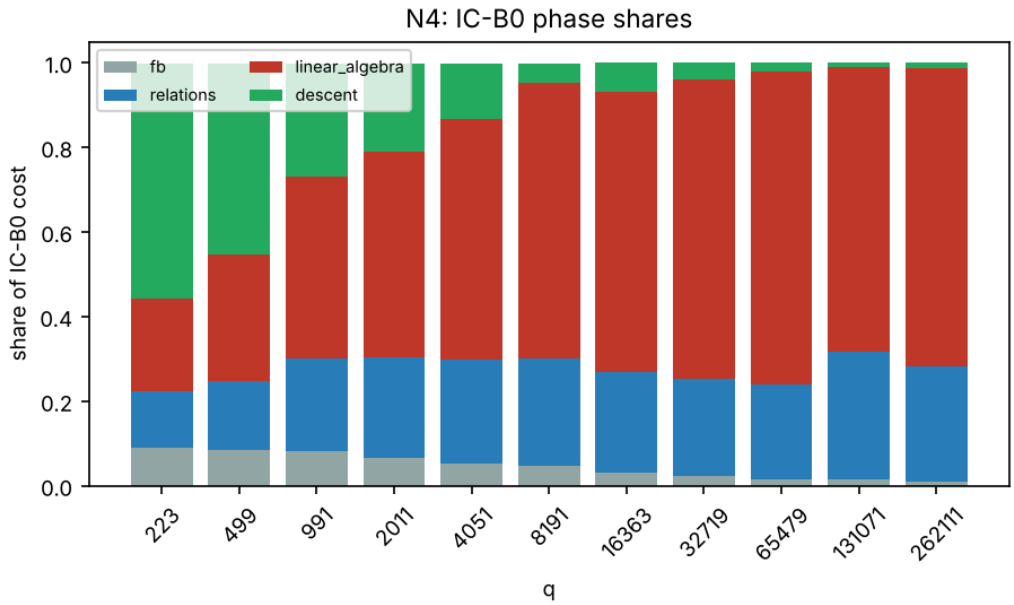


Figure 3: Phase shares of IC-B0 cost: linear algebra and relations dominate for q >= 4051; descent (the only phase that adds Pic classes) falls to 1%.

1.3. Exponents

range	IC-B0 (w=0)	rho (w=0)	gap (w=0)	gap (w=1)
all 11 sizes	0.998 [0.971, 1.023]	1.486 [1.464, 1.505]	0.487 [0.455, 0.521]	0.410 [0.378, 0.443]
q >= 4051 (primary)	1.083 [1.040, 1.129]	1.485 [1.437, 1.525]	0.402 [0.335, 0.468]	0.332 [0.266, 0.398]
IC-A0r, q <= 4051	1.986 [1.894, 2.072]	1.451 [1.407, 1.492]	−0.534 [−0.626, −0.422]	−0.554 [−0.643, −0.445]

1.4. T1: old quartic_ic.rs solves at q = 5, 7, 11

q	old S before targets	loop over n = 1..ell share	line relations (rank/ cols)	deg-3 pic_canon merges	old / canonical targets correct
5	650 443	99.82%	0 (0/6)	1	10/10, 5 (5 n/a)
7	2 751 342	99.95%	1 (1/8)	7	10/10, 7 (3 n/a)
11	37 920 004	99.99%	3 (3/12)	20	10/10, 10

The recovered scalars and the deg-6 table survive; the “recovery:ic” label, the deg-3 reduction, `2lp` (= `full`) and the rho S = 0 do not.

1.5. T3 and T4

Addition cost (3523M + 44I, 2227) vs published 148M + 15SQ + 2I (Flon, Oyono, Ritzenthaler, flex, char > 5; not implemented): descent share of IC-B0 is 55% at q = 223, 13% at 4051, 1% at 262111; with descent free the ratio at 4051 moves 0.735 to 0.637, at 262111 0.122 to 0.120. Not the bottleneck where IC wins; linear algebra is (57 to 74%). Stop rule: gap interval above 0 at w = 0 and w = 1, so N4 is open.

1.6. Limits

GHS-weak family (4 divides the order of E0, E0/F_{q^3}), plain rho without negation (sqrt 2 sensitivity), one shared 4-core VM at L0, wall time is a practicality note only (IC 0.57 of rho at q = 262111, count ratio 0.12), runs not bit-reproducible (HashMap order), arm A0 as frozen failed 5 of 15 units (A0r added post hoc), not compared with an external implementation.